Sumit Pathak

AI/ML and LLM Engineering

Mail: sumitpathak2904@gmail.com | Phone: 8076221099 | GitHub: Sumitpathak29 | LinkedIn: Sumitpathak29

Professional Summary

Final-year B.Tech Computer Science student at Symbiosis Institute of Technology, Pune, focused on **LLM Engineering, Generative AI, and Machine Learning**. Built practical AI systems involving **Retrieval-Augmented Generation (RAG), LoRA and QLoRA fine-tuning, Transformers, semantic search, and GPT architecture from scratch**, with a strong foundation in Python and PyTorch. Experienced in developing end-to-end AI applications spanning data processing, model development, retrieval, inference, and application integration. Seeking opportunities in **AI/ML, LLM Engineering, and Generative AI**.

Technical Skills

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Fine-Tuning, Semantic Search, Embeddings
LLM and Generative AI: RAG, Transformers, LoRA, QLoRA, Llama 3.2, LangChain, Ollama, Prompt Engineering
Languages: Python, C++, JavaScript, SQL, HTML, CSS
Vector Databases and Databases: ChromaDB, PostgreSQL, MongoDB
Backend and APIs: Node.js, Express.js, REST APIs, JWT Authentication
Frontend: React.js, HTML5, CSS3, Responsive Web Design
Tools and Cloud: Git, GitHub, Jupyter, AWS EC2, Gradio

Projects

Medical Research Assistant *Python, RAG, LangChain, ChromaDB, Llama 3.2, Ollama, Gradio*

- Built a **multi-document Retrieval-Augmented Generation (RAG) system** that processes medical PDFs, generates Sentence Transformer embeddings, stores document vectors in ChromaDB, and retrieves semantically relevant context for user queries.
- Implemented the RAG pipeline both **from scratch and with LangChain**, integrating Llama 3.2 through Ollama to generate context-grounded responses with document and page-level source attribution.
- Developed a Gradio-based interface for multi-PDF querying and explored **LoRA and QLoRA fine-tuning with 4-bit quantization** using Hugging Face Transformers, PEFT, and TRL.

GPT Language Model from Scratch *Python, PyTorch, NumPy*

- Implemented a **character-level, decoder-only GPT architecture from scratch** using PyTorch, including token and positional embeddings, multi-head causal self-attention, feed-forward networks, LayerNorm, and residual connections.
- Built the complete **autoregressive language-model training pipeline**, including character-level tokenization, mini-batch generation, cross-entropy loss, backpropagation, optimization, and next-token generation using the Tiny Shakespeare dataset.
- Implemented **causal attention masking** to prevent future-token information leakage and generated character-level text through autoregressive inference.

Movie Recommendation System *Python, Pandas, Scikit-learn, Machine Learning*

- Built a **content-based movie recommendation system** using the TMDB 5000 dataset to recommend movies based on similarity across movie attributes.
- Performed data preprocessing and feature engineering to construct meaningful movie representations and applied text-based similarity techniques to identify relevant recommendations.
- Developed an end-to-end recommendation pipeline that accepts a movie as input and returns a ranked list of similar movies based on feature similarity.

Education

Symbiosis Institute of Technology, Pune **2023–2027**
B.Tech in Computer Science and Engineering

Certifications

- **Nutanix Hybrid Cloud Fundamentals (NHCF)** — March 2026
- **Full-Stack Web Development Bootcamp** — Node.js, Express.js, React.js, Databases, REST APIs, Deployment
- **AI Engineering — LLM, RAG, QLoRA, Agents**